No: 8, First floor, 15th Cross, 100 Feet Ring Road, JP Nagar 6th Phase, Near Sarakki Signal, Bangalore – 560078, Phone: +91.80.42173649 Website: www.intepat.com, Email:contact@intepat.com

Patentability Search Report
On

"AI-BASED MULTI AGENT FRAMEWORK FOR REGIME- ADAPTIVE EQUITY RESEARCH"

Applicant: Satyam Das

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Understanding of the subject matter

The invention relates to an artificial intelligence-based equity research and investment analysis system and method designed to provide regime-adaptive stock evaluation, multi-perspective financial analysis, and investment decision support for securities markets. The system is configured to include a financial data ingestion layer, a market regime classification module, an investment scoring engine, a multi-agent analytical framework, a numerical reconciliation module, and a report generation platform that collectively enable automated and reliable equity research. The financial data ingestion layer is configured to retrieve live market data, company fundamentals, and other investment-related information from one or more data sources while generating provenance-tagged data records indicative of data reliability. The market regime classification module utilizes quantitative analysis techniques to identify prevailing market conditions and determine an appropriate market regime for investment evaluation. The investment scoring engine is configured to compute an investment readiness score using a plurality of financial factors and regime-dependent weighting models, while optionally incorporating market-specific indicators such as delivery percentage metrics for Indian equities. The multi-agent analytical framework comprises a plurality of specialist analyst agents configured to independently evaluate a target security from different analytical perspectives, including fundamental, technical, macroeconomic, sentiment, geopolitical, and adversarial viewpoints. The numerical reconciliation module is configured to verify and correct numerical claims generated by the analyst agents using authoritative financial data sources prior to synthesis. The system further incorporates a synthesis and reporting platform configured to generate consensus investment assessments, identify risks and opportunities, determine investment recommendations, and present structured investment reports. The overall objective of the invention is to provide an integrated AI-driven equity research platform that combines regime-adaptive scoring, multi-agent analysis, numerical verification, and consensus-based investment assessment within a unified framework, thereby enabling reliable and explainable investment decision support across multiple securities markets. The essential elements of the comprising:

- **Element 1(E1):** Receiving a target security and retrieving financial data from one or more data sources.
- **Element 1a (E1a):** Generating a provenance-tagged financial data context.
- **Element 1b (E1b):** Applying market-specific data retrieval and processing for different securities markets.
- **Element 2 (E2):** Classifying a market regime associated with the target security.
- **Element 3 (E3):** Computing an investment score using a plurality of financial factors and regime-dependent weighting.
- **Element 3a (E3a):** Using delivery percentage data as an indicator of investor conviction for Indian equities.
- **Element 3b (E3b):** Applying market-specific scoring and regime-classification models.
- **Element 4 (E4):** Dispatching a plurality of analyst agents to generate structured financial analyses.
- **Element 5 (E5):** Reconciling numerical claims generated by the analyst agents with authoritative financial data.
- **Element 6 (E6):** Generating an adversarial analysis to challenge an investment thesis.
- **Element 7 (E7):** Synthesizing the analyses to generate a consensus investment assessment.
- **Element 8 (E8):** Generating an investment analysis report comprising investment scores, verified financial information, and analytical conclusions.

Methodology of conducting the patent search

The prior art search has been conducted with respect to the invention disclosure given by **Applicant: Satyam Das** for its novelty and inventive step. Research is based on a set of identified keywords, International Patent Classification codes, references cited in relevant patent applications, and combinations of the above. The related patents were analysed against the essential elements set out at E1 to E8, and the conclusion at Section 10 is based on the consolidated analysis.

The Search Results section contains highly and moderately relevant cited prior art with full comment paragraphs and per-cell reasoning tables. The Other Interesting Results section contains additional patent references of comparatively lower relevance presented as table-line entries only. The Operational Platforms section contains operational platforms, commercial competitor products, open- source frameworks, standards-track non-patent literature, and academic references which an examiner may identify during broader searches, addressed in prose only without mapping table treatment.

Keywords

One or more of the keywords listed below have been used in different combinations while conducting the search:

Keywords
financial market data, equity market data, stock market data, financial data acquisition, live financial data
provenance-tagged financial data, verified financial data, data provenance, financial data verification
market-specific data processing, cross-market financial analysis, India US equity markets, multi-market financial data
numerical claim reconciliation, financial data validation, verified numerical values, authoritative financial data verification

adversarial AI agent, devil's advocate analysis, counter-thesis generation, adversarial investment analysis
consensus investment assessment, investment thesis generation, bull and bear analysis, investment risk assessment
AI-generated investment report, automated equity research, financial decision support, investment recommendation system
multi-agent financial analysis, AI analyst agents, parallel AI agents, collaborative investment analysis

Classification

International Patent Classification:

G06Q10/04 Forecasting or optimisation specially adapted for administrative or management purposes, e.g. linear programming or "cutting stock problem" (market predictions or forecasting for commercial activities **G06Q30/0202**)

G06Q40/06 Asset management; Financial planning or analysis

G06N20/20 Ensemble learning

G06Q30/02 Marketing; Price estimation or determination; Fundraising

G06Q10/06 Resources, workflows, human or project management; Enterprise or organisation planning; Enterprise or organisation modelling

G06Q40/00 Finance; Insurance; Tax strategies; Processing of corporate or income taxes

G06Q30/00 Commerce

G06Q40/04 Trading; Exchange, e.g. stocks, commodities, derivatives or currency exchange

G06N20/00 Machine learning

G06N7/00 Computing arrangements based on specific mathematical models

G06N3/04 Architecture, e.g. interconnection topology

G06N3/08 Learning methods

G06Q30/08 Auctions

G06F17/00 Digital computing or data processing equipment or methods, specially adapted for specific functions (information retrieval, database structures or file system structures therefor **G06F16/00**)

G06F40/56 Natural language generation

H04L51/02 using automatic reactions or user delegation, e.g. automatic replies or chatbot-generated messages

Sources used:

The databases and search engines used for conducting the patentability search are:

- ❑ Total Patent
- ❑ USPTO
- ❑ Espacenet
- ❑ WIPO
- ❑ PQAI
- ❑ Google Patents
- ❑ Patent Lens
- ❑ Google Scholar
- ❑ IEEE

Search Results:

Result 1

Patent Number	Inventor (s) / Assignee(s)	Date of Filing	Date of Publication / Grant
CN119295223A	Nantong Sankeitang Information Technology Co ltd	2024-11-01	2025-01-10
Title of the Patent	**An investment optimization system based on artificial intelligence investment research**		
Status	Withdrawn		

Abstract:

The invention discloses an investment optimizing system based on artificial intelligent investment research, which relates to the technical field of financial investment and comprises a data collecting module, a risk analyzing module and a risk analyzing module, wherein the data collecting module is used for collecting data related to investment combinations, including historical transaction data, market trend data and company financial data, preprocessing and storing the data, the risk analyzing module is used for analyzing the collected data, evaluating the risk level of the investment combinations and constructing a risk measurement prediction model, and the module can identify potential risk factors such as market fluctuation and industry risk and give corresponding risk scores and probability distribution.

Citations:

"An investment optimization system based on artificial intelligence investment research, which is characterized by comprising the following components: **The data collection module is responsible for collecting data related to the investment portfolio**, including historical transaction data, market trend data and company financial data, preprocessing and storing; **[Refer: Claim 1]**

The risk analysis module is used for analyzing the collected data, evaluating the risk level of the investment portfolio and constructing a risk measurement prediction model, and can identify potential risk factors such as market fluctuation and industry risk and give corresponding risk scores and probability distribution;

The optimization decision module is used for adjusting and optimizing the investment portfolio by utilizing an optimization algorithm **based on a risk analysis result, and can take factors such as risk preference, investment target and the like of investors into consideration to generate an optimized investment decision** scheme so as to reduce risks and improve return on investment; **[Refer: Claim 1]**

An investment optimization system based on artificial intelligence investment research as claimed in claim 1, wherein the data collection module comprises the specific steps of:

(1) Determining data sources to be collected, wherein the data sources comprise a financial database, a market data provider, a news report and a social media platform;

(2) Capturing relevant financial data from a determined data source by using a web crawler technology means;

(3) Data cleaning, namely cleaning the collected original data containing noise, repetition or irrelevant information by using an unsupervised learning algorithm, removing abnormal values, filling missing values and carrying out standardized treatment;

(4) Integrating the cleaned data according to a specific format or structure, so as to facilitate subsequent analysis and processing;

(5) Data storage-the integrated data is stored in a suitable database or data warehouse for subsequent data analysis and optimization. **[Refer: Claim 2]**

Perform risk assessment through the model output and present the risk assessment results to investors in the form of a report." **[Refer: Para 0076]**

Comment:

CN119295223A, hereinafter referred to as R1, relates to a financial investment evaluation system configured to collect financial information, evaluate financial products, predict risks, analyze assets, and provide investment decision support. R1 discloses a financial information acquisition module for collecting financial data, corresponding to Element (E1), and further discloses financial product evaluation using a classification model and scoring of financial indicators, corresponding to Elements (E2) and (E3). R1 additionally discloses asset analysis and investment evaluation functions for generating investment-related assessments, corresponding to Element (E8). R1 partially discloses Element (E4) by utilizing multiple evaluation and analysis modules for processing financial information; however, it does not disclose a plurality of specialist analyst agents generating structured analyses from distinct analytical perspectives. Further, R1 does not disclose provenance-tagged financial data (E1a), market-specific data processing (E1b), delivery percentage as an investor-conviction indicator (E3a), market-specific scoring and regime-classification models (E3b), numerical claim reconciliation with authoritative financial data (E5), adversarial analysis configured to challenge an investment thesis (E6), or synthesis of multiple agent outputs into a consensus investment assessment (E7). Accordingly, while R1 discloses financial data acquisition, investment scoring, asset evaluation, and investment decision support, it does not disclose the regime-adaptive scoring framework, multi-agent analytical architecture, numerical reconciliation mechanism, and adversarial investment analysis framework and therefore does not anticipate the subject matter of the present invention.

Result 2

Patent Number	Inventor (s) / Assignee(s)	Date of Filing	Date of Publication / Grant
US20220237700A1	Quantel Ai Inc	2022-01-18	2022-07-28
Title of the Patent	**Artificial intelligence investment platform**		
Status	Abandoned		

Abstract:

Systems and methods for an artificial intelligence (AI) platform operable to generate securities portfolio recommendations, including an AI engine operable to evaluate market sentiment data. The AI investment platform includes a plurality of AI engines, wherein each of the plurality of AI engines evaluates a distinct set of input factors. The AI investment platform is operable for autonomous operation using a plurality of learning techniques and/or predictive analytics techniques. The AI investment platform includes a social network.

Citations:

"In one embodiment, the present invention is directed to a portfolio management platform, including a server in network communication with a plurality of user devices, wherein the server is operable to generate a plurality of user profiles, **each associated with one or more of the plurality of user devices, wherein the plurality of user profiles include risk tolerance information** and desired returns over one or more time periods, wherein the server is in communication with a plurality of distinct artificial intelligence modules operable **to analyze data regarding one or more securities and generate recommendation data regarding the one or more securities, wherein the server generates a**

suggested portfolio securities allocation based on a weighted aggregation of the recommendation data generated by each of the plurality of distinct artificial intelligence modules and based on the risk tolerance information and the desired returns over the one or more time periods associated with the plurality of user profiles, wherein the weighted aggregation of the recommendation data is weighted based on historical data regarding the correlation of the recommendation data of each of the plurality of distinct artificial intelligence modules with previous performance data of each of the one or more securities, and **wherein the plurality of distinct artificial intelligence modules includes a sentiment analysis module configured to analyze sentiment data regarding the one or more securities. [Refer: Para 0018]**

Comment:

US20220237700A1, hereinafter referred to as R2, relates to an artificial intelligence-based investment platform configured to generate securities portfolio recommendations using a plurality of AI engines that evaluate different input factors and market sentiment data. R2 partially discloses receiving and processing financial and market-related information for investment analysis, corresponding to Element (E1), and further partially discloses market-specific processing by supporting analysis of securities and portfolio recommendations across investment environments, corresponding to Element (E1b). R2 additionally discloses a plurality of AI engines configured to independently evaluate distinct input factors, corresponding to Element (E4), and the aggregation of outputs from the AI engines to generate investment recommendations, corresponding to Element (E7). R2 partially discloses Element (E3) by evaluating multiple input factors for investment decision-making; however, it does not disclose computation of an investment score using regime-dependent weighting. Further, R2 does not disclose provenance-tagged financial data (E1a), market regime classification (E2), delivery percentage as an investor-conviction indicator for Indian equities (E3a), market-specific scoring and regime-classification models (E3b), numerical claim reconciliation with authoritative financial data (E5), or adversarial analysis configured to challenge an investment thesis (E6) and generating an investment analysis report (E8). Accordingly, while R2 discloses AI-based investment analysis, multiple AI engines, recommendation generation, and aggregation of analytical outputs, it does not disclose the regime-adaptive scoring framework, numerical reconciliation mechanism, provenance-tagged data architecture, and adversarial investment analysis framework and therefore does not anticipate the subject matter of the present invention.

Result 3

Patent Number	Inventor (s) / Assignee(s)	Date of Filing	Date of Publication / Grant
CN111754317A	Cao Mingzhou	2020-06-16	2020-10-09
Title of the Patent	**Financial investment data evaluation method and system**		
Status	Pending		

Abstract:

The invention belongs to the technical field of financial investment data evaluation, and discloses a financial investment data evaluation method and a financial investment data evaluation system, wherein the financial investment data evaluation system comprises: the system comprises a financial information acquisition module, a central control module, a financial product evaluation module, a financial risk prediction module, an asset analysis module, an investment evaluation module, a profit calculation module and a display module. According to the method, the financial product evaluation module is used for scoring the financial product target indexes by using the classification model, so that the classification effect and the calculation efficiency can be improved, the applicability of different financial product target indexes is expanded, and the accuracy of data in the evaluation process is improved; meanwhile, the reliability analysis of the financial assets according to the risk and income categories is realized through the asset analysis module, and decision support is provided for investors to carry out financial asset configuration.

Citations:

"A financial investment data evaluation system, comprising:

the system comprises a financial information acquisition module, a central control module, a financial product evaluation module, a financial risk prediction module, an asset analysis module, an investment evaluation module, a profit calculation module and a display module;

the financial information acquisition module is connected with the central control module and is used for acquiring financial information data;

the central control module is connected with the financial information acquisition module, the financial product evaluation module, the financial risk prediction module, the asset analysis module, the investment evaluation module, the income calculation module and the display module and is used for controlling the modules to normally work through the host; the financial product evaluation module is connected with the central control module and is used for evaluating the financial product through an evaluation program; the financial risk prediction module is connected with the central control module and used for predicting the financial investment risk through a prediction program; the asset analysis module is connected with the central control module and is used for analyzing the financial investment asset data through an analysis program; the investment evaluation module is connected with the central control module and is used for evaluating the financial investment data through an evaluation program; the income calculation module is connected with the central control module and is used for calculating financial investment income through a calculation program; and the display module is connected with the central control module and used for displaying the collected financial data, the evaluation result, the prediction result, the analysis result and the evaluation result through a display. **[Refer: Claim 1]**

The financial investment data evaluation method according to claim 1, wherein the financial investment data evaluation method comprises the steps of:

step one, acquiring financial information data through a financial information acquisition module; step two, the central control module evaluates the financial products by utilizing an evaluation program through a financial product evaluation module; predicting the financial investment risk by a financial risk prediction module by using a prediction program; analyzing the financial investment asset data by an asset analysis module by using an analysis program; evaluating the financial investment data by using an evaluation program through an investment evaluation module; calculating financial investment income by using a calculation program through an income calculation module; and fifthly, displaying the acquired financial data, the evaluation result, the prediction result, the analysis result and the evaluation result by using a display through a display module. **[Refer: Claim 2]**

The financial investment data evaluation system of claim 3, wherein the acquiring the first technical index data of the financial product for the first preset time after the preprocessing by the evaluation program comprises: acquiring first technical index data of the financial product within first preset time before preprocessing; and sequentially performing data conversion, data filling and data dimension reduction on the first technical index data of the financial product within the first preset time before the preprocessing to obtain the first technical index data of the financial product within the first preset time after the preprocessing. The financial investment data evaluation system of claim 4, wherein the sequentially performing data transformation, data population and data dimension reduction on the first technical index data of the financial product within a first preset time before the preprocessing comprises: binary conversion is carried out on the first technical index data of the financial product in the first preset time before the pretreatment by adopting a dummy vars function, so that the first technical index data of the financial product in the first preset time after the conversion is obtained; **filling missing values in the first technical index data of the converted financial product in the first preset time by adopting a pre-process function to obtain complete first technical index data of the financial product in the first preset time; calculating correlation between first technical index data of the financial product in a first preset time and correlation between the first technical index data and a financial product target index by adopting a cor function to obtain first correlation corresponding to the correlation between the first technical index data and second correlation corresponding to the correlation between the first technical index data and a dependent variable**; and if the first correlation degree of the index data in the first technical index data is greater than a first preset threshold value, and/or the second correlation degree between the index data in the first technical index data and the dependent variable is greater than a second preset threshold value, removing the index data in the first technical index data of the financial product in the complete first preset time. **[Refer: Claims 4-5]**

Comment:

CN111754317A, hereinafter referred to as R3, relates to a financial investment data evaluation system that acquires financial information, performs product evaluation / risk prediction / asset analysis / investment evaluation, computes user scores from technical indicators using a classification model, and displays the resulting outputs. R3 discloses acquiring financial information data and financial asset price-sequence data through a financial information acquisition / analysis workflow; evaluating financial products using technical indicator data processed through a trained classification model to generate a user score; analyzing asset data using risk-return indicators and clustering; and displaying acquired financial data together with evaluation, prediction, and analysis results. R3 maps to E1 (partially, financial information / asset data acquisition is disclosed but not receipt of a target security-specific workflow), E3 (partially, a score is generated from multiple technical / financial indicators, but no regime-dependent weighting is disclosed), E7 (partially, multiple analytical outputs are combined at a system level, but not as a consensus assessment from analyst analyses), and E8 (partially, display of financial data and evaluation / prediction / analysis outputs is disclosed). R3 is silent on E1a, E1b, E2, E3a, E3b, E4, E5, and E6; R3 most directly speaks to the integration of E2 with E3 and E4–E7 (regime-adaptive scoring integrated with a multi-agent, adversarial, and numeric-reconciliation investment-analysis architecture), and is silent on it. Therefore, R3 does not anticipate the present invention as a whole.

Result 4

Patent Number	**Inventor (s) / Assignee(s)**	**Date of Filing**	**Date of Publication / Grant**
US20190279301A1	*Dhruv Siddharth KRISHNAN*	*2019-05-06*	*2019-09-12*
Title of the Patent	**Systems and Methods Using an Algorithmic Solution for Analyzing a Portfolio of Stocks, Commodities and/or Other Financial Assets Based on Individual User Data to Achieve Desired Risk Based Financial Goals**		
Status	*Abandoned*		

Abstract:

A multi-factor qualitative method for assessing assets including receiving market data from a plurality of market data sources for processing by a trading app to make decisions as to whether to buy, sell, or hold a particular market asset; applying a data normalizer to the market data for processing dissimilar sets of

market data; receiving, by user input, a particular market asset in which a decision is required; processing, the market data by a complex event processing engine using an algorithmic solutions from a particular sector for market analysis of market data to generate a multi-factor model wherein the multi-factor model comprises: at least one of a set of a plurality of multi-factors related to the particular sector; applying, the multi-factor model by the trading app, to perform a multi-factor analysis making a decision whether to buy, sell or hold the market asset based on a normalized score.

Citations:

"A multi-factor qualitative method for assessing assets comprising:
receiving market data from a plurality of market data sources for processing by a trading app to make decisions as to whether to buy, sell, or hold a particular market asset; applying a data normalizer to the market data for processing dissimilar sets of market data; receiving, by user input, a particular market asset in which a decision to buy, sell or hold is required;
processing, the market data by a complex event processing engine using an algorithmic solutions from a particular sector for market analysis of market data to generate a multi-factor model wherein the multi-factor model comprises:
at least one of a set of a plurality of multi-factors related to the particular sector;
applying, the multi-factor model by the trading app, to perform a multi-factor analysis of the particular market asset and to generate a set of results; and
making a decision about the particular market asset by the trading app, from the set of results whether to buy, sell or hold the market asset based on a normalized score in the range of plus one to minus one of the set of results wherein a plus one score indicates a decision to buy and a negative one score indicates a decision to sell. **[Refer: Claim 1]**
The method of claim 1, further comprising: **applying the algorithmic solution for an information technologies market sector based on one or more factors comprising:**

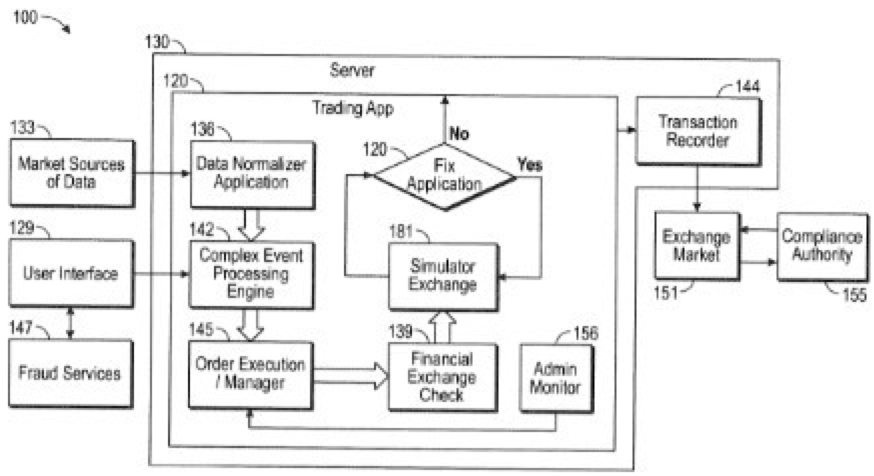
a difference in percentage of costs of software services compared to competitors; a difference in sales of products compared to competitors; and a difference in increased revenue compared to competitors.

The method of claim 1, further comprising: **applying the algorithmic solution for a telecommunication market sector based on one or more factors comprising**: a difference in percentage of costs for regulations added in previous years; a correlation of interest rates affecting market assets; a difference in growth compared to previous years of growth; and an amount of a change in a number of users for a particular related product or software services to the market asset. **[Refer: Claims 6-7]**

The method of claim 1, further comprising: creating, by the trading app, a user profile for each user wherein the user profile is an additional factor used in the multi-factor model for making a decision whether to buy, sell or hold the particular market asset **[Refer: Claim 2]**

The method of claim 1, further comprising: applying the algorithmic solution for an information technologies market sector based on one or more factors comprising: a difference in percentage of costs of software services compared to competitors; a difference in sales of products compared to competitors; and a difference in increased revenue compared to competitors. [Refer: Claim 6]

Figures:



Comment:

US20190279301A1, hereinafter referred to as R4, relates to a multi-factor trading application that receives market data from multiple sources, normalizes dissimilar data, applies sector-specific factor models, and generates a normalized buy/sell/hold decision for a selected market asset. R4 discloses receipt of market data from a plurality of market data sources and receipt of a particular market asset for analysis; application of a data normalizer and a multi-factor model to generate results and a normalized buy/sell/hold score; and sector-specific factor models for information-technology and telecommunication assets. R4 maps to E1 (partially, target asset input and multi-source market-data retrieval), E3 (partially, multi-factor scoring/decision generation), E1b (partially, only to the extent of sector-specific processing rather than different securities-market processing), E3b (partially, only to the extent of sector-specific scoring models rather than market-specific scoring and regime-classification models), and E7 (partially, combining factor-analysis results into a final asset decision). R4 is silent on E1a, E2, E3a, E4, E5, E6, and E8. R4 most directly speaks to the integration of E2 with E4/E5/E6/E7 (i.e., a regime-adaptive scoring pipeline combined with multi-agent analysis, numeric reconciliation, adversarial analysis, and consensus synthesis), and is silent on it. Therefore, R4 does not anticipate the present invention.

Result 5

Patent Number	**Inventor (s) / Assignee(s)**	**Date of Filing**	**Date of Publication / Grant**
US11410240B2	*Lopez De Prado Lopez Marcos M*	*2020-09-10*	*2022-08-09*
Title of the Patent	***Tactical investment algorithms through Monte Carlo backtesting***		
Status	*Active*		

Abstract:
Disclosed herein are methods and systems for computing optimized trading instructions, comprising applying a plurality of prediction models to compute trading instructions predicted to produce optimal outcomes based on received trading observations. Each prediction model is optimized, using a Monte Carlo algorithm, to predict optimal outcomes for a respective trading pattern based on a respective Data Generating Processes (DGPs). Weights are assigned to the prediction models according to probability scores computed for them according to mapping of the received trading observations to the DGPs associated with the prediction models and aggregated trading instructions are computed by aggregating, based on the weights, the trading instructions computed by the prediction models. Further disclosed are systems and methods for creating an ensemble of prediction models each optimized to predict optimal outcomes for a respective one of a plurality of trading patterns using an expanded simulated samples' dataset generated by a Monte Carlo algorithm.

Citations:

"A computer implemented method of improving quality of computed optimized trading instructions, comprising:

generating noise reduced and expended training dataset comprising a plurality of simulated trading observations, by: **accessing at least one external storage device and/or at least one network storage resource;**

fetching from said at least one external storage device and/or said at least one network storage resource a plurality of historical datasets comprising a plurality of past trading observations reflecting a respective one of a plurality of trading patterns; deriving from said plurality of past trading observations a plurality of Data Generating Processes (DGPs); and generating said noise reduced and expanded training dataset by executing a Monte Carlo algorithm based on a respective one of the plurality of DGPs; **receiving a dataset comprising a plurality of trading observations; accessing a prediction models repository to fetch a plurality of prediction models; increasing a quality of computed trading instructions by:** applying said plurality of prediction models to compute trading instructions to produce optimal outcomes predicted based on the plurality of trading observations, each of the plurality of prediction models is optimized to predict optimal outcomes for a respective one of the plurality of trading patterns by training the respective model through backtesting using the expanded training dataset;

computing a respective probability score for mapping the received dataset to each of the plurality of DGPs; assigning a respective weight to each of the plurality of prediction models according to its respective probability score; and

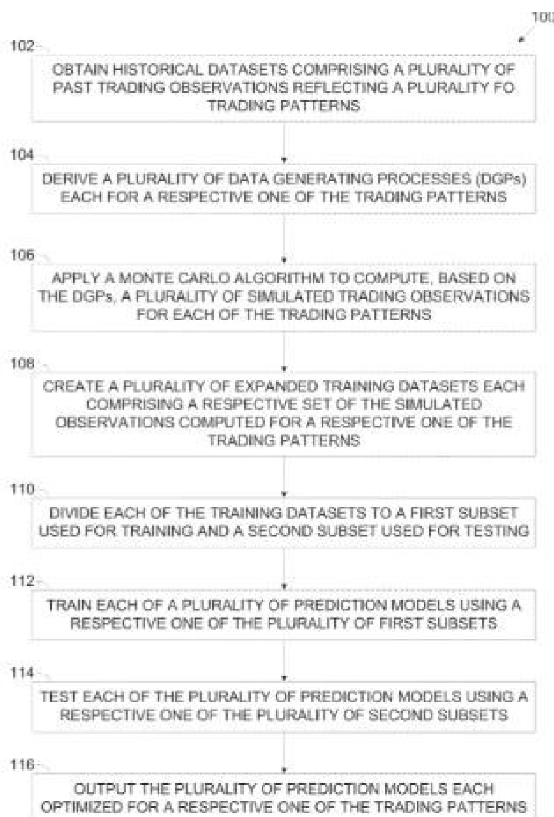
computing aggregated trading instructions aggregating the trading instructions for investing a respective portion of an investment according to each of the plurality of prediction models based on its respective weight; and

outputting the aggregated trading instructions to at least one trading system. **[Refer: Claim 1]**

The computer implemented method of claim 1, **wherein the plurality of trading observations comprise a plurality of trading features observed during a**

certain recent time period associated with respective outcomes which jointly reflect a current trading pattern. [Refer: Claim 2] The computer implemented method of claim 1, further comprising computing the **respective probability score for mapping the received dataset to each of the plurality of DGPs based on a comparison between the respective outcomes extracted from the plurality of trading observations and the optimal outcomes computed by each of the plurality of prediction models. [Refer: Claim 4]**

Figures:



Comment:

US11410240B2, hereinafter referred to as R5, relates to a pattern-adaptive ensemble trading system that maps current trading observations to historical trading patterns/DGPs and weights multiple prediction models to generate aggregated trading instructions. R5 discloses retrieval of historical trading datasets and receipt of trading observations; derivation of DGPs for trading patterns; computation of probability scores mapping current data to DGPs; weighting of prediction models based on those scores; and aggregation of model outputs into trading instructions. R5 maps to E1 (partially, retrieval of trading/market datasets rather than a target security-specific financial research input), E2 (partially, mapping current observations to trading patterns/DGPs rather than classifying a market regime of a target security), E3 (partially, weighted multi-model trading output rather than factor-based investment scoring), and E7 (partially, aggregation of model outputs into trading instructions). R5 is silent on E1a, E1b, E3a, E3b, E4, E5, E6, and E8. R5 most directly speaks to the integration of E2 with E3 (pattern-dependent weighting of multiple prediction models for aggregated trading instructions), and is silent on the core combination of regime-adaptive equity scoring with multi-agent analyst generation, adversarial analysis, and numeric reconciliation; Therefore R5 does not anticipate the subject matter of the present invention.

Result 6

Patent Number	Inventor (s) / Assignee(s)	Date of Filing	Date of Publication / Grant
CN110490746A	*Fujian University of Technology*	*2019-08-19*	*2019-11-22*
Title of the Patent	***Fintech artificial intelligence optimization method and device for quantitative investment in stock market***		
Status	*Pending*		

Abstract:

"The invention discloses a kind of financial technology artificial intelligence optimization's methods of stock market quantization investment. Including obtaining stock historical price data, stock historical price data are pre-processed, data after being handled；Data after processing are imported in the optimal neural network model that screening obtains, obtains Prediction of Stock Index price; Prediction of Stock Index price is directed into default rule, optimal Portfolio ratio is obtained. The invention also discloses a kind of financial technology artificial intelligence optimization's devices of stock market quantization investment. Financial investment combinatorial optimization algorithm proposed by the present invention structure neural network based, adaptive ability, simulation nonlinear function feature, it is predicted using financial technology analysis stock certificate data, function of the innovation artificial intelligence optimization's algorithm in security quantization investment combination."

Citations:

“A fintech artificial intelligence optimization method for quantitative investment in the stock market, characterized by comprising: **Obtaining historical stock price data, performing preprocessing on the historical stock price data, and obtaining processed data;** Importing the processed data into the optimal neural network model obtained by screening to obtain the stock forecast price;

The stock forecast price is imported into the preset rules to obtain the optimal investment portfolio ratio. **[Refer: Claim 1]**

The financial technology artificial intelligence optimization method of a kind of stock market quantitative investment according to claim 1, characterized in that **"preprocessing the historical stock price data" specifically includes:**

Read the csv file of the stock historical price data, save the csv file with DataFrame type and delete the rows containing null values, and obtain the usable stock historical data; Divide the available stock historical data into training data and test data according to a preset ratio; The training data and the test data are averaged and normalized. **[Refer: Claim 2]**

Comment:

CN110490746A,hereinafter referred to as R6, relates to relates to an AI-based quantitative stock-investment method for forecasting stock prices from historical stock price data and deriving an optimal portfolio ratio. R6 discloses obtaining and preprocessing historical stock price data, and using the processed stock data for stock-price prediction and portfolio-ratio determination. R6 maps to E1 (partially, limited to obtaining historical stock price data rather than receiving a target security and retrieving broader financial data from one or more data sources). R6 is silent on E1a, E1b, E2, E3, E3a, E3b, E4, E5, E6, E7, and E8. R6 most directly speaks to the integration of E2 with E3 (regime-based investment scoring using market-state-dependent weighting), and is silent on it. Therefore R6 does not anticipate the subject matter of the present invention.

Result 7

Patent Number	**Inventor (s) / Assignee(s)**	**Date of Filing**	**Date of Publication / Grant**
US12086561B2	International Business Machines Corp	2019-08-27	2021-03-04
Title of the Patent	***Multi-agent conversational agent framework***		
Status	Active		

Abstract:

One embodiment provides a method, including: receiving, within a conversational agent creation framework used to create conversational agents that perform negotiations on behalf of users, (i) a selection of one of a plurality of conversational agent roles (ii) negotiation constraints for the selected role, and (iii) an object for the negotiation; creating a conversational agent (i) having the selected role and (ii) programmed with the negotiation constraints; identifying at least one other conversational agent (i) having an opposing role with respect to the object and (ii) being programmed with opposing role negotiation constraints; and conducting a conversational session between the conversational agent and the at least one other conversational agent, wherein the conversational session comprises the negotiation for the object and wherein the conversational agent and the at least one other conversational agent perform the negotiation in view of the negotiation constraints and the opposing role negotiation constraints.

Citations:

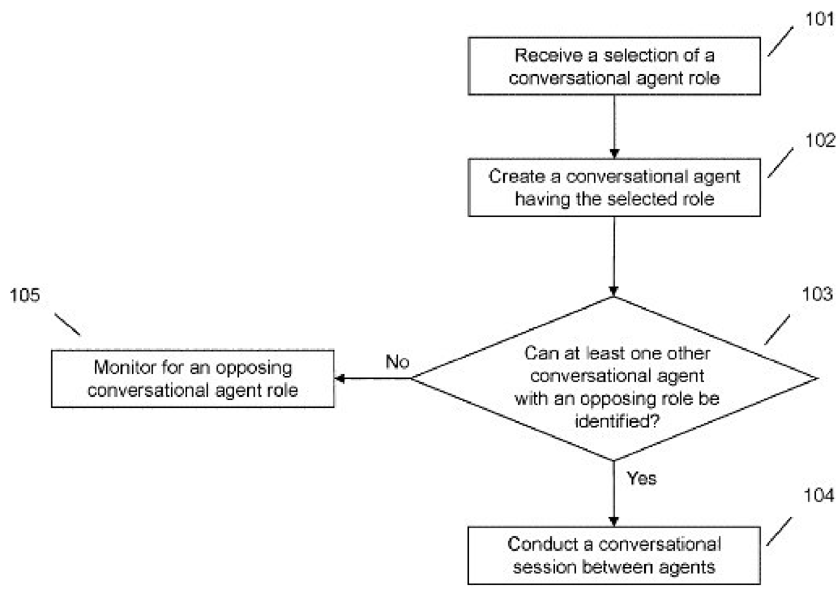
A method, comprising: providing **a conversational agent creation framework that creates conversational agents based upon input from a plurality of users, wherein within the conversational agent** creation framework the conversational agents interact and perform negotiations on behalf of users for objects without requiring input from the plurality of users; receiving, within the conversational agent creation framework and from one of the plurality of users, input to generate an instance of a conversational agent to perform a negotiation on behalf of the at least one of the plurality of users, wherein the input comprises (i) a selection of one of a plurality of conversational agent roles, each of the plurality of conversational agent roles corresponding to a role within a negotiation, (ii) negotiation constraints for the selected role, and (iii) an object for the negotiation, wherein the conversational agent creation framework comprises a plurality of created instances of a conversational agent having a particular role, wherein the instance is unique to the one of the plurality of users the object identified by the user, wherein the conversational agent creation framework comprises a plurality of created instances of a conversational agent having a particular role; creating, within the conversational agent creation framework, the instance of the conversational agent, wherein the instance (i) has the selected role, (ii) is programmed with the negotiation constraints, and (iii) is unique to the object for the negotiation and the one of the plurality of users;

identifying, within the conversational agent creation framework, at least one other instance of a conversational agent, wherein the at least one other instance (i) has an opposing role with respect to the object and (ii) is programmed with opposing role negotiation constraints;

conducting, within a negotiation instance created within the conversational agent creation framework, the negotiation instance being unique to the instance of the conversational agent, the at least one other instance of a conversational agent, and the object, in response to identifying the at least one other instance of a conversational agent, a conversational session between a conversational agent corresponding to the instance of a conversational agent and at least one other conversational agent corresponding to the at least one other instance of a conversational agent, wherein the conducting comprises each of the conversational agent and the at least one other conversational agent traversing nodes included within a dialog tree generated based upon a negotiation specification comprising data corresponding to the negotiation instance, the roles of the conversational agent and the at least one other conversational agent, and the negotiation constraints;**[Refer: Claim 1]**

...... the instance of the conversational agent, and the at least one other instance of a conversational agent, wherein each of the negotiation entities comprise object identifiers for the object of a corresponding negotiation entity; and providing, **from the conversational agent creation framework and responsive to completing a negotiation, information to the one of the plurality of users** regarding the completed negotiation." **[Refer: Claim 1]**

***Figures*:**



Comment:

US12086561B2, hereinafter referred to as R7, relates to a conversational agent framework configured to create multiple agents having different negotiation roles and conduct negotiations between agents with opposing objectives. R7 partially discloses a plurality of conversational agents with distinct roles, corresponding to Element (E4), and further discloses an opposing conversational agent for conducting counter-negotiations, corresponding to Element (E6). R7 also partially discloses interaction between multiple agents to reach a negotiation outcome, corresponding to Element (E7). However, R7 does not disclose receiving and retrieving financial data (E1), generating a provenance-tagged financial data context (E1a), applying market-specific data retrieval and processing (E1b), classifying a market regime (E2), computing an investment score using financial factors and regime-dependent weighting (E3), using delivery percentage as an investor-conviction indicator (E3a), applying market-specific scoring and regime-classification models (E3b), reconciling numerical claims with authoritative financial data (E5), or generating an investment analysis report (E8). Accordingly, while R7 discloses a multi-agent negotiation framework with opposing agents, it does not disclose the financial analysis and investment evaluation framework of the present invention and therefore does not anticipate the subject matter of the present invention.

<u>Other Interesting Results:</u>

Number and Title	Applicant	Date of publication/ Grant
WO2006044858A2 *System and method for analyzing analyst recommendations on a single stock basis*	*Starmine Corp*	*2007-04-26*
WO2017015672A1 *Topological data analysis for identification of market regimes for prediction*	*Ayasdi Inc*	*2018-01-23*
CN110400222A *A kind of Stock Market Forecasting method based on hidden Markov and deep learning*	*South China University of Technology SCUT; South China Normal University*	*2019-11-01*
US12086561B2 *Multi-agent conversational agent framework*	*International Business Machines Corp*	*2024-09-10*
CN115358193A *Financial information generating method, device, equipment and medium*	*China Merchants Bank Co Ltd*	*2022-11-18*

Operational Platforms, Frameworks, and Non-Patent Technical References

The references below relate to AI-based investment platforms, financial data evaluation systems, equity scoring methods, multi-factor investment models, multi-agent financial analysis frameworks, reinforcement learning-based investment systems, conversational AI architectures, and financial decision support technologies. They are discussed because an examiner may identify them during broader searches relating to financial data processing, equity research, investment scoring, market regime analysis, multi-agent artificial intelligence, financial forecasting, portfolio recommendation systems, and automated investment decision-making. While these references disclose various aspects of financial data acquisition, investment scoring, AI-based financial analysis, multi-agent architectures, and investment recommendation techniques, none appears to disclose the specific combination of provenance-tagged financial data, regime-adaptive investment scoring, market-specific processing, numerical claim reconciliation, adversarial multi-agent analysis, and consensus-based investment assessment within a unified AI-driven equity research framework as described in the present invention.

Multi-Agent Reinforcement Learning for Financial Investment

Multi-Agent Reinforcement Learning for Financial Investment relates to a multi-agent reinforcement learning-based financial investment decision system configured to generate investment strategies using reinforcement learning, Hidden Markov Models (HMM), and cooperative multi-agent interaction. This reference partially discloses receiving and processing financial investment data, corresponding to Element (E1), and discloses classifying financial market conditions using a Hidden Markov Model, corresponding to Element (E2). This reference further partially discloses computation of investment decisions based on financial indicators, corresponding to Element (E3), and discloses a plurality of cooperative agents for investment decision-making, corresponding to Element

(E4). This reference also partially discloses synthesis of agent outputs through cooperative learning to determine an optimal investment decision, corresponding to Element (E7). However, this reference does not disclose generating a provenance-tagged financial data context (E1a), applying market-specific data retrieval and processing (E1b), using delivery percentage as an investor-conviction indicator (E3a), applying market-specific scoring and regime-classification models (E3b), reconciling numerical claims with authoritative financial data (E5), generating an adversarial analysis (E6), or generating an investment analysis report comprising verified financial information and analytical conclusions (E8). Accordingly, while this reference discloses HMM-based market analysis and a cooperative multi-agent investment framework, it does not disclose the provenance-based verification, numerical reconciliation, adversarial analysis, and regime-adaptive equity research framework of the present invention and therefore does not anticipate the subject matter of the present invention.

A Multi-Agent System for Equity Analysis

A Multi-Agent System for Equity Analysis relates to a multi-agent artificial intelligence architecture configured to support equity analysis by integrating structured market data and unstructured financial documents through a plurality of specialized AI agents coordinated by a manager agent. This reference partially discloses receiving and processing financial market data, corresponding to Element (E1), and discloses a plurality of specialized agents configured to perform different analytical functions, corresponding to Element (E4). This reference further discloses integrating outputs generated by the specialized agents to support investment analysis, corresponding to Element (E7), and partially discloses generation of analytical results for equity research and decision support, corresponding to Element (E8). However, this reference does not disclose generating a provenance-tagged financial data context (E1a), applying market-specific data retrieval and processing (E1b), classifying a market regime (E2), computing an investment score using financial factors and regime-dependent weighting (E3), using delivery percentage as an investor-conviction indicator

(E3a), applying market-specific scoring and regime-classification models (E3b), reconciling numerical claims with authoritative financial data (E5), or generating an adversarial analysis configured to challenge an investment thesis (E6). Accordingly, while this reference discloses a multi-agent AI framework for equity analysis, it does not disclose the regime-adaptive scoring, numerical reconciliation, provenance-based verification, and adversarial analysis framework of the present invention and therefore does not anticipate the subject matter of the present invention.

AI Applications in the Chinese Financial Market

AI Applications in the Chinese Financial Market relates to the application of artificial intelligence in financial markets for improving financial decision-making, risk prediction, and regulatory efficiency. This reference partially discloses the use of financial market data for AI-based analysis, corresponding to Element (E1), and further partially discloses providing AI-assisted financial insights for decision-making, corresponding to Element (E8). However, this reference does not disclose generating a provenance-tagged financial data context (E1a), applying market-specific data retrieval and processing (E1b), classifying a market regime (E2), computing an investment score using financial factors and regime-dependent weighting (E3), using delivery percentage as an investor-conviction indicator (E3a), applying market-specific scoring and regime-classification models (E3b), dispatching a plurality of analyst agents (E4), reconciling numerical claims with authoritative financial data (E5), generating an adversarial analysis (E6), or synthesizing multiple analyses into a consensus investment assessment (E7). Accordingly, while this reference generally discloses AI-assisted financial analysis, it does not disclose the regime-adaptive, multi-agent, and numerical reconciliation framework of the present invention and therefore does not anticipate the subject matter of the present invention.

CONCLUSION

The cited prior art collectively discloses various AI-based investment platforms, financial data evaluation systems, equity scoring models, multi-factor investment frameworks, multi-agent financial analysis architectures, reinforcement learning-based investment approaches, conversational agent frameworks, and financial decision support technologies. R1 is the closest reference with respect to financial data acquisition, investment evaluation, and investment scoring. R2 is the closest reference regarding AI-based investment analysis, multiple AI engines, and investment recommendation generation. R3 discloses multi-factor investment evaluation and weighted investment scoring methodologies. R4 is the closest reference regarding Hidden Markov Model-based market analysis, multi-agent reinforcement learning, and collaborative investment decision-making. R5 discloses AI-assisted financial analysis and predictive decision support. R6 is the closest reference regarding specialized multi-agent AI architecture for equity analysis and synthesis of analytical outputs. R7 is the closest reference regarding multi-agent interaction, opposing conversational agents, and adversarial reasoning. However, none of the cited references, either alone or in combination, discloses the complete architecture of the present invention comprising provenance-tagged financial data, market-specific data processing, regime-adaptive investment scoring, numerical claim reconciliation with authoritative financial data, adversarial multi-agent analysis, and consensus-based investment assessment within a unified AI-driven equity research framework. In light of the above analysis, the above-cited prior art does not disclose all the essential features of the present invention in combination. Therefore, close prior art is discovered but does not impact the patentability of the invention. The letter 'P' denotes the partial mapping of an element with respect to the cited prior art and the symbol '✓' denotes complete mapping of an element with respect to the cited prior art.

S. No	Prior Art	Elements											
		E1	E1 a	E1 b	E2	E3	E3 a	E3 b	E4	E5	E6	E7	E8
1.	**CN119295223A**	P	X	X	✓	✓	X	X	P	X	X	X	✓
2.	**US20220237700A1**	P	X	P	X	P	X	X	✓	X	X	✓	X
3.	**CN111754317A**	P	X	X	X	P	X	X	X	X	X	P	P
4.	**US20190279301A1**	P	X	P	X	P	X	P	X	X	X	P	X
5.	**US11410240B2**	P	X	X	P	P	X	X	X	X	X	P	X
6.	**CN110490746A**	P	X	X	X	X	X	X	X	X	X	X	X
7.	**US12086561B2**	X	X	X	X	X	X	X	P	X	✓	P	X

On a scale of 1 to 5, the strength of the invention may be **2**

Hence, the features of the present Invention Disclosure may have a moderate probability of getting a patent grant.

Note: The scale mentioned above is as follows:

Scale	Definition
1	No close prior art discovered.
2	Close prior art discovered, but do not impact on the patentability of the invention.
3	Close prior art discovered, but it is possible to differentiate the invention from the prior art, in terms of the patentability criteria.
4	Very close prior art discovered and differences between the prior arts and the invention are trivial and may not be patentable.
5	Very close prior art found with the prior art(s) hitting right on the invention.

DISCLAIMER

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